

National Land Acquisition & Management System

Smart India Hackathon (SIH) 2026 Technical Report

MEMBER PROFILE: Member 1

Technical Role: Frontend & Decision Dashboard Developer

Primary Focus: React State & Hooks Management (useState, useEffect for API data binding), JWT Client-Side Storage

1. Functional Domain & Work Done

As part of the NLAMS development team, Member 1 has taken ownership of the functional module targeting Frontend & Decision Dashboard Developer. The following components have been fully implemented and integrated into the live workspace:

- **Saffron/Navy/Green Light Theme Portal:** Refactored the landing pages and dashboard to match a formal Indian Government light-theme visual language, removing dark components.
- **Dynamic React Language Switcher:** Created translation maps and hooked Navbar globe selector to immediately swap English texts to Hindi for all forms, charts, and buttons.
- **Recharts & GIS Integration:** Bound live SQLite stats aggregates into the dashboard cards and dynamic leaflet map polygons.

2. Overall Website Layout & Features

The NLAMS portal operates as a single source of truth under a modernised, high-contrast, official government light theme (saffron, navy, and emerald accents). Key features include:

- **Tricolor Home & Directives:** An entry portal showcasing the LARR Act 2013 roadmap and operational pillars.
- **Bilingual Language Switcher:** A global toggle translating all KPI labels, timelines, Gazette notifications, and Treasury advice receipts dynamically into Hindi and English.
- **Executive Dashboard & GIS:** Renders land parcels as Leaflet map overlay polygons, color-coded by acquisition stage, alongside Recharts predictive dashboards.
- **Automated LARR Workflows:** A pipeline allowing Central, State, and District roles to sign off on stages, instantly generating printable official Gazette templates.
- **Web3 Trust & Compensation Portal:** Simulates smart contract escrow releases with Metamask wallet connectivity, a digital deed SHA-256 validation scanner, and a scrolling block ledger.
- **Mobile-Responsive Field Survey Node:** Allows field surveyors to capture coordinate coordinates using GPS triangulation and report soil classification, structures, and site images.
- **SMS Alert Simulator:** Fires dynamic, floating mobile network text notification alerts during proposal submissions, boundary checks, and DBT escrow releases.

3. Sequential Learning Pathway & Project Roadmap

To ensure a smooth transition from prototype to production for the SIH finals, the following learning benchmarks are mapped for this role:

- **Focus Area:** React State & Hooks Management (useState, useEffect for API data binding)
- **Focus Area:** JWT Client-Side Storage & Request Authorization headers injection
- **Focus Area:** Conditional Role-Based Dashboard layout rendering (District, Surveyor, Ministry)
- **Focus Area:** Dynamic charts implementation using Recharts SVG nodes

4. Judge Defense Recommendations

During the pitch and technical Q&A, the judges will test the integrity of each module. Use the following defense arguments to demonstrate technical authority:

- **Defense Tip:** State that the UI-based role filters are for operator convenience and workflow simulation, while the actual API routes are secured backend-side through cryptographic checks and middleware.